

not required.) However, notice that the total mole number of the reactants (2 kmol) is not equal to the total mole number of the products (1 kmol). That is, *the total number of moles is not conserved during a chemical reaction.*

A frequently used quantity in the analysis of combustion processes to quantify the amounts of fuel and air is the **air–fuel ratio** AF. It is usually expressed on a mass basis and is defined as *the ratio of the mass of air to the mass of fuel* for a combustion process (Fig. 15–6). That is,

$$AF = \frac{m_{\text{air}}}{m_{\text{fuel}}} \quad (15-3)$$

The mass m of a substance is related to the number of moles N through the relation $m = NM$, where M is the molar mass.

The air–fuel ratio can also be expressed on a mole basis as the ratio of the mole numbers of air to the mole numbers of fuel. But we will use the former definition. The reciprocal of air–fuel ratio is called the **fuel–air ratio**.

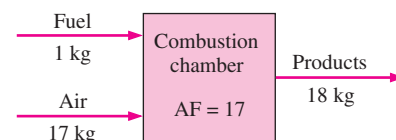


FIGURE 15–6

The air–fuel ratio (AF) represents the amount of air used per unit mass of fuel during a combustion process.

EXAMPLE 15–1 Balancing the Combustion Equation

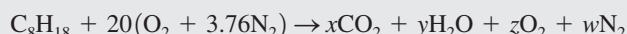
One kmol of octane (C_8H_{18}) is burned with air that contains 20 kmol of O_2 , as shown in Fig. 15–7. Assuming the products contain only CO_2 , H_2O , O_2 , and N_2 , determine the mole number of each gas in the products and the air–fuel ratio for this combustion process.

Solution The amount of fuel and the amount of oxygen in the air are given. The amount of the products and the AF are to be determined.

Assumptions The combustion products contain CO_2 , H_2O , O_2 , and N_2 only.

Properties The molar mass of air is $M_{\text{air}} = 28.97 \text{ kg/kmol} \cong 29.0 \text{ kg/kmol}$ (Table A–1).

Analysis The chemical equation for this combustion process can be written as



where the terms in the parentheses represent the composition of dry air that contains 1 kmol of O_2 and x , y , z , and w represent the unknown mole numbers of the gases in the products. These unknowns are determined by applying the mass balance to each of the elements—that is, by requiring that the total mass or mole number of each element in the reactants be equal to that in the products:

$$\begin{array}{lll} \text{C:} & 8 = x & \rightarrow x = \mathbf{8} \\ \text{H:} & 18 = 2y & \rightarrow y = \mathbf{9} \\ \text{O:} & 20 \times 2 = 2x + y + 2z & \rightarrow z = \mathbf{7.5} \\ \text{N}_2: & (20)(3.76) = w & \rightarrow w = \mathbf{75.2} \end{array}$$

Substituting yields



Note that the coefficient 20 in the balanced equation above represents the number of moles of *oxygen*, not the number of moles of air. The latter is obtained by adding $20 \times 3.76 = 75.2$ moles of nitrogen to the 20 moles of

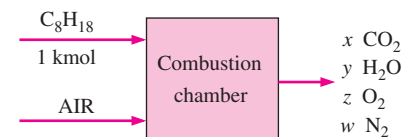


FIGURE 15–7

Schematic for Example 15–1.

oxygen, giving a total of 95.2 moles of air. The air–fuel ratio (AF) is determined from Eq. 15–3 by taking the ratio of the mass of the air and the mass of the fuel,

$$\begin{aligned} \text{AF} &= \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(NM)_{\text{air}}}{(NM)_{\text{C}} + (NM)_{\text{H}_2}} \\ &= \frac{(20 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(8 \text{ kmol})(12 \text{ kg/kmol}) + (9 \text{ kmol})(2 \text{ kg/kmol})} \\ &= \mathbf{24.2 \text{ kg air/kg fuel}} \end{aligned}$$

That is, 24.2 kg of air is used to burn each kilogram of fuel during this combustion process.

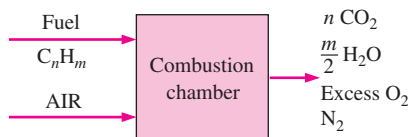


FIGURE 15–8

A combustion process is complete if all the combustible components of the fuel are burned to completion.

☐	
☐	
	$\text{CH}_4 + 2(\text{O}_2 + 3.76\text{N}_2) \rightarrow$
	$\text{CO}_2 + 2\text{H}_2\text{O} + 7.52\text{N}_2$
	• no unburned fuel
	• no free oxygen in products
☐	

FIGURE 15–9

The complete combustion process with no free oxygen in the products is called theoretical combustion.

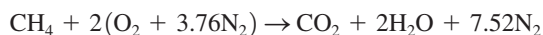
15–2 ■ THEORETICAL AND ACTUAL COMBUSTION PROCESSES

It is often instructive to study the combustion of a fuel by assuming that the combustion is complete. A combustion process is **complete** if all the carbon in the fuel burns to CO_2 , all the hydrogen burns to H_2O , and all the sulfur (if any) burns to SO_2 . That is, all the combustible components of a fuel are burned to completion during a complete combustion process (Fig. 15–8). Conversely, the combustion process is **incomplete** if the combustion products contain any unburned fuel or components such as C, H_2 , CO, or OH.

Insufficient oxygen is an obvious reason for incomplete combustion, but it is not the only one. Incomplete combustion occurs even when more oxygen is present in the combustion chamber than is needed for complete combustion. This may be attributed to insufficient mixing in the combustion chamber during the limited time that the fuel and the oxygen are in contact. Another cause of incomplete combustion is *dissociation*, which becomes important at high temperatures.

Oxygen has a much greater tendency to combine with hydrogen than it does with carbon. Therefore, the hydrogen in the fuel normally burns to completion, forming H_2O , even when there is less oxygen than needed for complete combustion. Some of the carbon, however, ends up as CO or just as plain C particles (soot) in the products.

The minimum amount of air needed for the complete combustion of a fuel is called the **stoichiometric** or **theoretical air**. Thus, when a fuel is completely burned with theoretical air, no uncombined oxygen is present in the product gases. The theoretical air is also referred to as the *chemically correct amount of air*, or *100 percent theoretical air*. A combustion process with less than the theoretical air is bound to be incomplete. The ideal combustion process during which a fuel is burned completely with theoretical air is called the **stoichiometric** or **theoretical combustion** of that fuel (Fig. 15–9). For example, the theoretical combustion of methane is



Notice that the products of the theoretical combustion contain no unburned methane and no C, H_2 , CO, OH, or free O_2 .

In actual combustion processes, it is common practice to use more air than the stoichiometric amount to increase the chances of complete combustion or to control the temperature of the combustion chamber. The amount of air in excess of the stoichiometric amount is called **excess air**. The amount of excess air is usually expressed in terms of the stoichiometric air as **percent excess air** or **percent theoretical air**. For example, 50 percent excess air is equivalent to 150 percent theoretical air, and 200 percent excess air is equivalent to 300 percent theoretical air. Of course, the stoichiometric air can be expressed as 0 percent excess air or 100 percent theoretical air. Amounts of air less than the stoichiometric amount are called **deficiency of air** and are often expressed as **percent deficiency of air**. For example, 90 percent theoretical air is equivalent to 10 percent deficiency of air. The amount of air used in combustion processes is also expressed in terms of the **equivalence ratio**, which is the ratio of the actual fuel–air ratio to the stoichiometric fuel–air ratio.

Predicting the composition of the products is relatively easy when the combustion process is assumed to be complete and the exact amounts of the fuel and air used are known. All one needs to do in this case is simply apply the mass balance to each element that appears in the combustion equation, without needing to take any measurements. Things are not so simple, however, when one is dealing with actual combustion processes. For one thing, actual combustion processes are hardly ever complete, even in the presence of excess air. Therefore, it is impossible to predict the composition of the products on the basis of the mass balance alone. Then the only alternative we have is to measure the amount of each component in the products directly.

A commonly used device to analyze the composition of combustion gases is the **Orsat gas analyzer**. In this device, a sample of the combustion gases is collected and cooled to room temperature and pressure, at which point its volume is measured. The sample is then brought into contact with a chemical that absorbs the CO_2 . The remaining gases are returned to the room temperature and pressure, and the new volume they occupy is measured. The ratio of the reduction in volume to the original volume is the volume fraction of the CO_2 , which is equivalent to the mole fraction if ideal-gas behavior is assumed (Fig. 15–10). The volume fractions of the other gases are determined by repeating this procedure. In Orsat analysis the gas sample is collected over water and is maintained saturated at all times. Therefore, the vapor pressure of water remains constant during the entire test. For this reason the presence of water vapor in the test chamber is ignored and data are reported on a dry basis. However, the amount of H_2O formed during combustion is easily determined by balancing the combustion equation.

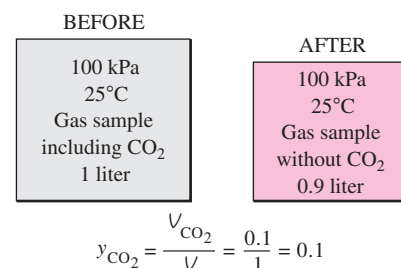


FIGURE 15–10

Determining the mole fraction of the CO_2 in combustion gases by using the Orsat gas analyzer.

EXAMPLE 15–2 Dew-Point Temperature of Combustion Products

Ethane (C_2H_6) is burned with 20 percent excess air during a combustion process, as shown in Fig. 15–11. Assuming complete combustion and a total pressure of 100 kPa, determine (a) the air–fuel ratio and (b) the dew-point temperature of the products.

Solution The fuel is burned completely with excess air. The AF and the dew point of the products are to be determined.

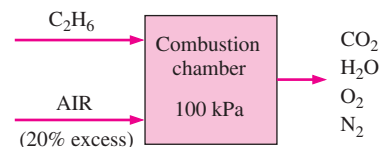
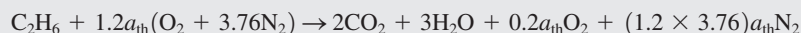


FIGURE 15–11

Schematic for Example 15–2.

Assumptions 1 Combustion is complete. 2 Combustion gases are ideal gases.

Analysis The combustion products contain CO_2 , H_2O , N_2 , and some excess O_2 only. Then the combustion equation can be written as



where a_{th} is the stoichiometric coefficient for air. We have automatically accounted for the 20 percent excess air by using the factor $1.2a_{\text{th}}$ instead of a_{th} for air. The stoichiometric amount of oxygen ($a_{\text{th}}\text{O}_2$) is used to oxidize the fuel, and the remaining excess amount ($0.2a_{\text{th}}\text{O}_2$) appears in the products as unused oxygen. Notice that the coefficient of N_2 is the same on both sides of the equation, and that we wrote the C and H balances directly since they are so obvious. The coefficient a_{th} is determined from the O_2 balance to be

$$\text{O}_2: \quad 1.2a_{\text{th}} = 2 + 1.5 + 0.2a_{\text{th}} \rightarrow a_{\text{th}} = 3.5$$

Substituting,



(a) The air–fuel ratio is determined from Eq. 15–3 by taking the ratio of the mass of the air to the mass of the fuel,

$$\begin{aligned} \text{AF} &= \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(4.2 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(2 \text{ kmol})(12 \text{ kg/kmol}) + (3 \text{ kmol})(2 \text{ kg/kmol})} \\ &= \mathbf{19.3 \text{ kg air/kg fuel}} \end{aligned}$$

That is, 19.3 kg of air is supplied for each kilogram of fuel during this combustion process.

(b) The dew-point temperature of the products is the temperature at which the water vapor in the products starts to condense as the products are cooled at constant pressure. Recall from Chap. 14 that the dew-point temperature of a gas–vapor mixture is the saturation temperature of the water vapor corresponding to its partial pressure. Therefore, we need to determine the partial pressure of the water vapor P_v in the products first. Assuming ideal-gas behavior for the combustion gases, we have

$$P_v = \left(\frac{N_v}{N_{\text{prod}}} \right) (P_{\text{prod}}) = \left(\frac{3 \text{ kmol}}{21.49 \text{ kmol}} \right) (100 \text{ kPa}) = 13.96 \text{ kPa}$$

Thus,

$$T_{\text{dp}} = T_{\text{sat @ 13.96 kPa}} = \mathbf{52.3^\circ\text{C}} \quad (\text{Table A-5})$$

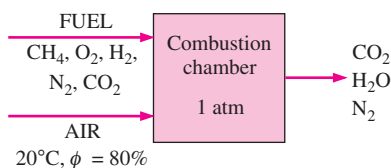


FIGURE 15-12

Schematic for Example 15–3.

EXAMPLE 15-3 Combustion of a Gaseous Fuel with Moist Air

A certain natural gas has the following volumetric analysis: 72 percent CH_4 , 9 percent H_2 , 14 percent N_2 , 2 percent O_2 , and 3 percent CO_2 . This gas is now burned with the stoichiometric amount of air that enters the combustion chamber at 20°C , 1 atm, and 80 percent relative humidity, as shown in Fig. 15–12. Assuming complete combustion and a total pressure of 1 atm, determine the dew-point temperature of the products.

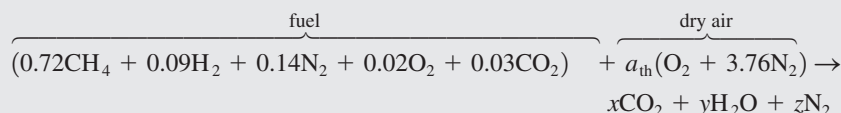
Solution A gaseous fuel is burned with the stoichiometric amount of moist air. The dew point temperature of the products is to be determined.

Assumptions 1 The fuel is burned completely and thus all the carbon in the fuel burns to CO_2 and all the hydrogen to H_2O . 2 The fuel is burned with the stoichiometric amount of air and thus there is no free O_2 in the product gases. 3 Combustion gases are ideal gases.

Properties The saturation pressure of water at 20°C is 2.3392 kPa (Table A-4).

Analysis We note that the moisture in the air does not react with anything; it simply shows up as additional H_2O in the products. Therefore, for simplicity, we balance the combustion equation by using dry air and then add the moisture later to both sides of the equation.

Considering 1 kmol of fuel,



The unknown coefficients in the above equation are determined from mass balances on various elements,

$$\text{C:} \quad 0.72 + 0.03 = x \rightarrow x = 0.75$$

$$\text{H:} \quad 0.72 \times 4 + 0.09 \times 2 = 2y \rightarrow y = 1.53$$

$$\text{O}_2: \quad 0.02 + 0.03 + a_{\text{th}} = x + \frac{y}{2} \rightarrow a_{\text{th}} = 1.465$$

$$\text{N}_2: \quad 0.14 + 3.76a_{\text{th}} = z \rightarrow z = 5.648$$

Next we determine the amount of moisture that accompanies $4.76a_{\text{th}} = (4.76)(1.465) = 6.97$ kmol of dry air. The partial pressure of the moisture in the air is

$$P_{v,\text{air}} = \phi_{\text{air}} P_{\text{sat @ } 20^\circ\text{C}} = (0.80)(2.3392 \text{ kPa}) = 1.871 \text{ kPa}$$

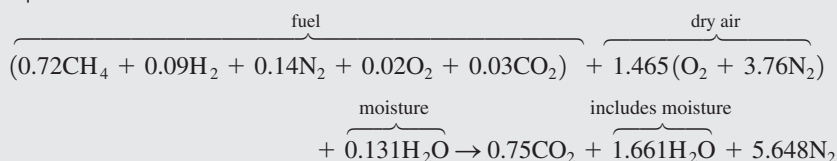
Assuming ideal-gas behavior, the number of moles of the moisture in the air is

$$N_{v,\text{air}} = \left(\frac{P_{v,\text{air}}}{P_{\text{total}}} \right) N_{\text{total}} = \left(\frac{1.871 \text{ kPa}}{101.325 \text{ kPa}} \right) (6.97 + N_{v,\text{air}})$$

which yields

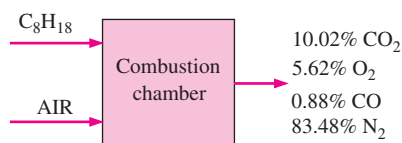
$$N_{v,\text{air}} = 0.131 \text{ kmol}$$

The balanced combustion equation is obtained by substituting the coefficients determined earlier and adding 0.131 kmol of H_2O to both sides of the equation:



The dew-point temperature of the products is the temperature at which the water vapor in the products starts to condense as the products are cooled. Again, assuming ideal-gas behavior, the partial pressure of the water vapor in the combustion gases is

$$P_{v,\text{prod}} = \left(\frac{N_{v,\text{prod}}}{N_{\text{prod}}} \right) P_{\text{prod}} = \left(\frac{1.661 \text{ kmol}}{8.059 \text{ kmol}} \right) (101.325 \text{ kPa}) = 20.88 \text{ kPa}$$

**FIGURE 15-13**

Schematic for Example 15-4.

Thus,

$$T_{\text{dp}} = T_{\text{sat @ 20.88 kPa}} = \mathbf{60.9^{\circ}\text{C}}$$

Discussion If the combustion process were achieved with dry air instead of moist air, the products would contain less moisture, and the dew-point temperature in this case would be 59.5°C.

EXAMPLE 15-4 Reverse Combustion Analysis

Octane (C_8H_{18}) is burned with dry air. The volumetric analysis of the products on a dry basis is (Fig. 15-13)

CO_2 :	10.02 percent
O_2 :	5.62 percent
CO :	0.88 percent
N_2 :	83.48 percent

Determine (a) the air–fuel ratio, (b) the percentage of theoretical air used, and (c) the amount of H_2O that condenses as the products are cooled to 25°C at 100 kPa.

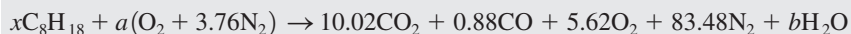
Solution Combustion products whose composition is given are cooled to 25°C. The AF, the percent theoretical air used, and the fraction of water vapor that condenses are to be determined.

Assumptions Combustion gases are ideal gases.

Properties The saturation pressure of water at 25°C is 3.1698 kPa (Table A-4).

Analysis Note that we know the relative composition of the products, but we do not know how much fuel or air is used during the combustion process. However, they can be determined from mass balances. The H_2O in the combustion gases will start condensing when the temperature drops to the dew-point temperature.

For ideal gases, the volume fractions are equivalent to the mole fractions. Considering 100 kmol of dry products for convenience, the combustion equation can be written as



The unknown coefficients x , a , and b are determined from mass balances,

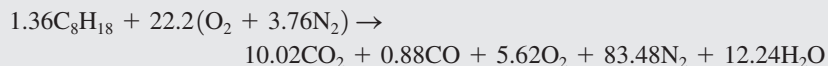
$$\text{N}_2: \quad 3.76a = 83.48 \quad \rightarrow \quad a = 22.20$$

$$\text{C}: \quad 8x = 10.02 + 0.88 \quad \rightarrow \quad x = 1.36$$

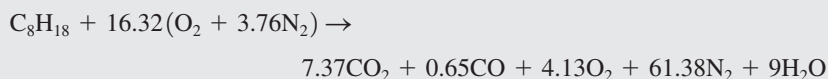
$$\text{H}: \quad 18x = 2b \quad \rightarrow \quad b = 12.24$$

$$\text{O}_2: \quad a = 10.02 + 0.44 + 5.62 + \frac{b}{2} \quad \rightarrow \quad 22.20 = 22.20$$

The O_2 balance is not necessary, but it can be used to check the values obtained from the other mass balances, as we did previously. Substituting, we get



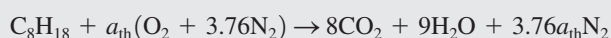
The combustion equation for 1 kmol of fuel is obtained by dividing the above equation by 1.36,



(a) The air–fuel ratio is determined by taking the ratio of the mass of the air to the mass of the fuel (Eq. 15–3),

$$\begin{aligned}\text{AF} &= \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(16.32 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(8 \text{ kmol})(12 \text{ kg/kmol}) + (9 \text{ kmol})(2 \text{ kg/kmol})} \\ &= \mathbf{19.76 \text{ kg air/kg fuel}}\end{aligned}$$

(b) To find the percentage of theoretical air used, we need to know the theoretical amount of air, which is determined from the theoretical combustion equation of the fuel,



$$\text{O}_2: \quad a_{\text{th}} = 8 + 4.5 \rightarrow a_{\text{th}} = 12.5$$

Then,

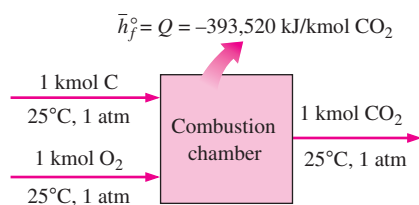
$$\begin{aligned}\text{Percentage of theoretical air} &= \frac{m_{\text{air,act}}}{m_{\text{air,th}}} = \frac{N_{\text{air,act}}}{N_{\text{air,th}}} \\ &= \frac{(16.32)(4.76) \text{ kmol}}{(12.50)(4.76) \text{ kmol}} \\ &= \mathbf{131\%}\end{aligned}$$

That is, 31 percent excess air was used during this combustion process. Notice that some carbon formed carbon monoxide even though there was considerably more oxygen than needed for complete combustion.

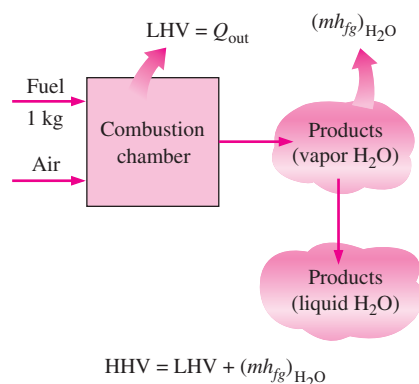
(c) For each kmol of fuel burned, $7.37 + 0.65 + 4.13 + 61.38 + 9 = 82.53$ kmol of products are formed, including 9 kmol of H_2O . Assuming that the dew-point temperature of the products is above 25°C , some of the water vapor will condense as the products are cooled to 25°C . If N_w kmol of H_2O condenses, there will be $(9 - N_w)$ kmol of water vapor left in the products. The mole number of the products in the gas phase will also decrease to $82.53 - N_w$ as a result. By treating the product gases (including the remaining water vapor) as ideal gases, N_w is determined by equating the mole fraction of the water vapor to its pressure fraction,

$$\begin{aligned}\frac{N_v}{N_{\text{prod,gas}}} &= \frac{P_v}{P_{\text{prod}}} \\ \frac{9 - N_w}{82.53 - N_w} &= \frac{3.1698 \text{ kPa}}{100 \text{ kPa}} \\ N_w &= \mathbf{6.59 \text{ kmol}}\end{aligned}$$

Therefore, the majority of the water vapor in the products (73 percent of it) condenses as the product gases are cooled to 25°C .

**FIGURE 15-18**

The enthalpy of formation of a compound represents the amount of energy absorbed or released as the component is formed from its stable elements during a steady-flow process at a specified state.

**FIGURE 15-19**

The higher heating value of a fuel is equal to the sum of the lower heating value of the fuel and the latent heat of vaporization of the H_2O in the products.

Now reconsider the formation of CO_2 (a compound) from its elements C and O_2 at 25°C and 1 atm during a steady-flow process. The enthalpy change during this process was determined to be $-393,520 \text{ kJ/kmol}$. However, $H_{\text{react}} = 0$ since both reactants are elements at the standard reference state, and the products consist of 1 kmol of CO_2 at the same state. Therefore, the enthalpy of formation of CO_2 at the standard reference state is $-393,520 \text{ kJ/kmol}$ (Fig. 15–18). That is,

$$\bar{h}_{f,\text{CO}_2}^o = -393,520 \text{ kJ/kmol}$$

The negative sign is due to the fact that the enthalpy of 1 kmol of CO_2 at 25°C and 1 atm is $393,520 \text{ kJ}$ less than the enthalpy of 1 kmol of C and 1 kmol of O_2 at the same state. In other words, $393,520 \text{ kJ}$ of chemical energy is released (leaving the system as heat) when C and O_2 combine to form 1 kmol of CO_2 . Therefore, a negative enthalpy of formation for a compound indicates that heat is released during the formation of that compound from its stable elements. A positive value indicates heat is absorbed.

You will notice that two $\bar{h}_f^o$ values are given for H_2O in Table A–26, one for liquid water and the other for water vapor. This is because both phases of H_2O are encountered at 25°C , and the effect of pressure on the enthalpy of formation is small. (Note that under equilibrium conditions, water exists only as a liquid at 25°C and 1 atm.) The difference between the two enthalpies of formation is equal to the h_{fg} of water at 25°C , which is 2441.7 kJ/kg or $44,000 \text{ kJ/kmol}$.

Another term commonly used in conjunction with the combustion of fuels is the **heating value** of the fuel, which is defined as the amount of heat released when a fuel is burned completely in a steady-flow process and the products are returned to the state of the reactants. In other words, the heating value of a fuel is equal to the absolute value of the enthalpy of combustion of the fuel. That is,

$$\text{Heating value} = |h_c| \quad (\text{kJ/kg fuel})$$

The heating value depends on the *phase* of the H_2O in the products. The heating value is called the **higher heating value** (HHV) when the H_2O in the products is in the liquid form, and it is called the **lower heating value** (LHV) when the H_2O in the products is in the vapor form (Fig. 15–19). The two heating values are related by

$$\text{HHV} = \text{LHV} + (mh_{fg})_{\text{H}_2\text{O}} \quad (\text{kJ/kg fuel}) \quad (15-7)$$

where m is the mass of H_2O in the products per unit mass of fuel and h_{fg} is the enthalpy of vaporization of water at the specified temperature. Higher and lower heating values of common fuels are given in Table A–27.

The heating value or enthalpy of combustion of a fuel can be determined from a knowledge of the enthalpy of formation for the compounds involved. This is illustrated with the following example.

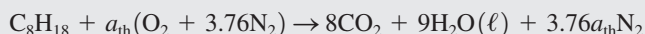
EXAMPLE 15-5 Evaluation of the Enthalpy of Combustion

Determine the enthalpy of combustion of liquid octane (C_8H_{18}) at 25°C and 1 atm, using enthalpy-of-formation data from Table A–26. Assume the water in the products is in the liquid form.

Solution The enthalpy of combustion of a fuel is to be determined using enthalpy of formation data.

Properties The enthalpy of formation at 25°C and 1 atm is $-393,520$ kJ/kmol for CO_2 , $-285,830$ kJ/kmol for $\text{H}_2\text{O}(\ell)$, and $-249,950$ kJ/kmol for $\text{C}_8\text{H}_{18}(\ell)$ (Table A-26).

Analysis The combustion of C_8H_{18} is illustrated in Fig. 15-20. The stoichiometric equation for this reaction is



Both the reactants and the products are at the standard reference state of 25°C and 1 atm. Also, N_2 and O_2 are stable elements, and thus their enthalpy of formation is zero. Then the enthalpy of combustion of C_8H_{18} becomes (Eq. 15-6)

$$\begin{aligned}\bar{h}_C &= H_{\text{prod}} - H_{\text{react}} \\ &= \sum N_p \bar{h}_{f,p}^\circ - \sum N_r \bar{h}_{f,r}^\circ = (N\bar{h}_f^\circ)_{\text{CO}_2} + (N\bar{h}_f^\circ)_{\text{H}_2\text{O}} - (N\bar{h}_f^\circ)_{\text{C}_8\text{H}_{18}}\end{aligned}$$

Substituting,

$$\begin{aligned}\bar{h}_C &= (8 \text{ kmol})(-393,520 \text{ kJ/kmol}) + (9 \text{ kmol})(-285,830 \text{ kJ/kmol}) \\ &\quad - (1 \text{ kmol})(-249,950 \text{ kJ/kmol}) \\ &= \mathbf{-5,471,000 \text{ kJ/kmol C}_8\text{H}_{18} = -47,891 \text{ kJ/kg C}_8\text{H}_{18}}\end{aligned}$$

which is practically identical to the listed value of 47,890 kJ/kg in Table A-27. Since the water in the products is assumed to be in the liquid phase, this h_C value corresponds to the HHV of liquid C_8H_{18} .

Discussion It can be shown that the result for gaseous octane is $-5,512,200$ kJ/kmol or $-48,255$ kJ/kg.

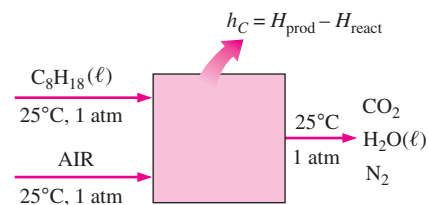


FIGURE 15-20

Schematic for Example 15-5.

When the exact composition of the fuel is known, the *enthalpy of combustion* of that fuel can be determined using enthalpy of formation data as shown above. However, for fuels that exhibit considerable variations in composition depending on the source, such as coal, natural gas, and fuel oil, it is more practical to determine their enthalpy of combustion experimentally by burning them directly in a bomb calorimeter at constant volume or in a steady-flow device.

15-4 ■ FIRST-LAW ANALYSIS OF REACTING SYSTEMS

The energy balance (or the first-law) relations developed in Chaps. 4 and 5 are applicable to both reacting and nonreacting systems. However, chemically reacting systems involve changes in their chemical energy, and thus it is more convenient to rewrite the energy balance relations so that the changes in chemical energies are explicitly expressed. We do this first for steady-flow systems and then for closed systems.

Steady-Flow Systems

Before writing the energy balance relation, we need to express the enthalpy of a component in a form suitable for use for reacting systems. That is, we need to express the enthalpy such that it is relative to the standard reference

A combustion chamber normally involves heat output but no heat input. Then the energy balance for a *typical steady-flow combustion process* becomes

$$Q_{\text{out}} = \underbrace{\sum N_r(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_r}_{\text{Energy in by mass per mole of fuel}} - \underbrace{\sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p}_{\text{Energy out by mass per mole of fuel}} \quad (15-13)$$

It expresses that the heat output during a combustion process is simply the difference between the energy of the reactants entering and the energy of the products leaving the combustion chamber.

Closed Systems

The general closed-system energy balance relation $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ can be expressed for a stationary *chemically reacting closed system* as

$$(Q_{\text{in}} - Q_{\text{out}}) + (W_{\text{in}} - W_{\text{out}}) = U_{\text{prod}} - U_{\text{react}} \quad (\text{kJ/kmol fuel}) \quad (15-14)$$

where U_{prod} represents the internal energy of the products and U_{react} represents the internal energy of the reactants. To avoid using another property—the *internal energy of formation* $\bar{u}_f^\circ$ —we utilize the definition of enthalpy ($\bar{u} = \bar{h} - P\bar{v}$ or $\bar{u}_f^\circ + \bar{u} - \bar{u}^\circ = \bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v}$) and express the above equation as (Fig. 15-22)

$$Q - W = \sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v})_p - \sum N_r(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v})_r \quad (15-15)$$

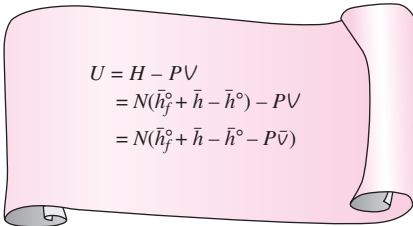
where we have taken heat transfer *to* the system and work done *by* the system to be *positive* quantities. The $P\bar{v}$ terms are negligible for solids and liquids, and can be replaced by $R_u T$ for gases that behave as an ideal gas. Also, if desired, the $\bar{h} - P\bar{v}$ terms in Eq. 15-15 can be replaced by $\bar{u}$.

The work term in Eq. 15-15 represents all forms of work, including the boundary work. It was shown in Chap. 4 that $\Delta U + W_b = \Delta H$ for nonreacting closed systems undergoing a quasi-equilibrium $P = \text{constant}$ expansion or compression process. This is also the case for chemically reacting systems.

There are several important considerations in the analysis of reacting systems. For example, we need to know whether the fuel is a solid, a liquid, or a gas since the enthalpy of formation h_f° of a fuel depends on the phase of the fuel. We also need to know the state of the fuel when it enters the combustion chamber in order to determine its enthalpy. For entropy calculations it is especially important to know if the fuel and air enter the combustion chamber premixed or separately. When the combustion products are cooled to low temperatures, we need to consider the possibility of condensation of some of the water vapor in the product gases.

EXAMPLE 15-6 First-Law Analysis of Steady-Flow Combustion

Liquid propane (C_3H_8) enters a combustion chamber at 25°C at a rate of 0.05 kg/min where it is mixed and burned with 50 percent excess air that enters the combustion chamber at 7°C , as shown in Fig. 15-23. An analysis of the combustion gases reveals that all the hydrogen in the fuel burns to H_2O but only 90 percent of the carbon burns to CO_2 , with the remaining 10 percent forming CO . If the exit temperature of the combustion gases is



$$\begin{aligned} U &= H - PV \\ &= N(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ) - PV \\ &= N(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v}) \end{aligned}$$

FIGURE 15-22

An expression for the internal energy of a chemical component in terms of the enthalpy.

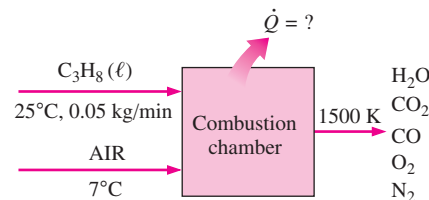


FIGURE 15-23

Schematic for Example 15-6.

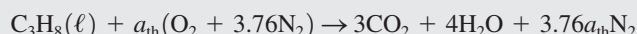
1500 K, determine (a) the mass flow rate of air and (b) the rate of heat transfer from the combustion chamber.

Solution Liquid propane is burned steadily with excess air. The mass flow rate of air and the rate of heat transfer are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air and the combustion gases are ideal gases. **3** Kinetic and potential energies are negligible.

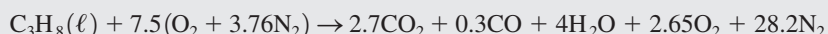
Analysis We note that all the hydrogen in the fuel burns to H_2O but 10 percent of the carbon burns incompletely and forms CO. Also, the fuel is burned with excess air and thus there is some free O_2 in the product gases.

The theoretical amount of air is determined from the stoichiometric reaction to be



$$\text{O}_2 \text{ balance:} \quad a_{\text{th}} = 3 + 2 = 5$$

Then the balanced equation for the actual combustion process with 50 percent excess air and some CO in the products becomes



(a) The air–fuel ratio for this combustion process is

$$\begin{aligned} \text{AF} &= \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(7.5 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(3 \text{ kmol})(12 \text{ kg/kmol}) + (4 \text{ kmol})(2 \text{ kg/kmol})} \\ &= 25.53 \text{ kg air/kg fuel} \end{aligned}$$

Thus,

$$\begin{aligned} \dot{m}_{\text{air}} &= (\text{AF})(\dot{m}_{\text{fuel}}) \\ &= (25.53 \text{ kg air/kg fuel})(0.05 \text{ kg fuel/min}) \\ &= \mathbf{1.18 \text{ kg air/min}} \end{aligned}$$

(b) The heat transfer for this steady-flow combustion process is determined from the steady-flow energy balance $E_{\text{out}} = E_{\text{in}}$ applied on the combustion chamber per unit mole of the fuel,

$$Q_{\text{out}} + \sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p = \sum N_r(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_r$$

or

$$Q_{\text{out}} = \sum N_r(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_r - \sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p$$

Assuming the air and the combustion products to be ideal gases, we have $h = h(T)$, and we form the following minitable using data from the property tables:

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{280 \text{ K}}$ kJ/kmol	$\bar{h}_{298 \text{ K}}$ kJ/kmol	$\bar{h}_{1500 \text{ K}}$ kJ/kmol
$\text{C}_3\text{H}_8(\ell)$	−118,910	—	—	—
O_2	0	8150	8682	49,292
N_2	0	8141	8669	47,073
$\text{H}_2\text{O}(g)$	−241,820	—	9904	57,999
CO_2	−393,520	—	9364	71,078
CO	−110,530	—	8669	47,517

The $\bar{h}_f^\circ$ of liquid propane is obtained by subtracting the $\bar{h}_{fg}$ of propane at 25°C from the $\bar{h}_f^\circ$ of gas propane. Substituting gives

$$\begin{aligned} \dot{Q}_{\text{out}} &= (1 \text{ kmol C}_3\text{H}_8)[(-118,910 + h_{298} - h_{298}) \text{ kJ/kmol C}_3\text{H}_8] \\ &\quad + (7.5 \text{ kmol O}_2)[(0 + 8150 - 8682) \text{ kJ/kmol O}_2] \\ &\quad + (28.2 \text{ kmol N}_2)[(0 + 8141 - 8669) \text{ kJ/kmol N}_2] \\ &\quad - (2.7 \text{ kmol CO}_2)[(-393,520 + 71,078 - 9364) \text{ kJ/kmol CO}_2] \\ &\quad - (0.3 \text{ kmol CO})[(-110,530 + 47,517 - 8669) \text{ kJ/kmol CO}] \\ &\quad - (4 \text{ kmol H}_2\text{O})[(-241,820 + 57,999 - 9904) \text{ kJ/kmol H}_2\text{O}] \\ &\quad - (2.65 \text{ kmol O}_2)[(0 + 49,292 - 8682) \text{ kJ/kmol O}_2] \\ &\quad - (28.2 \text{ kmol N}_2)[(0 + 47,073 - 8669) \text{ kJ/kmol N}_2] \\ &= 363,880 \text{ kJ/kmol of C}_3\text{H}_8 \end{aligned}$$

Thus 363,880 kJ of heat is transferred from the combustion chamber for each kmol (44 kg) of propane. This corresponds to $363,880/44 = 8270$ kJ of heat loss per kilogram of propane. Then the rate of heat transfer for a mass flow rate of 0.05 kg/min for the propane becomes

$$\dot{Q}_{\text{out}} = \dot{m}_{\text{out}} = (0.05 \text{ kg/min})(8270 \text{ kJ/kg}) = 413.5 \text{ kJ/min} = \mathbf{6.89 \text{ kW}}$$

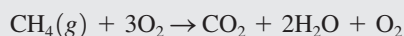
EXAMPLE 15-7 First-Law Analysis of Combustion in a Bomb

The constant-volume tank shown in Fig. 15-24 contains 1 lbmol of methane (CH_4) gas and 3 lbmol of O_2 at 77°F and 1 atm. The contents of the tank are ignited, and the methane gas burns completely. If the final temperature is 1800 R, determine (a) the final pressure in the tank and (b) the heat transfer during this process.

Solution Methane is burned in a rigid tank. The final pressure in the tank and the heat transfer are to be determined.

Assumptions **1** The fuel is burned completely and thus all the carbon in the fuel burns to CO_2 and all the hydrogen to H_2O . **2** The fuel, the air, and the combustion gases are ideal gases. **3** Kinetic and potential energies are negligible. **4** There are no work interactions involved.

Analysis The balanced combustion equation is



(a) At 1800 R, water exists in the gas phase. Using the ideal-gas relation for both the reactants and the products, the final pressure in the tank is determined to be

$$\left. \begin{aligned} P_{\text{react}} V &= N_{\text{react}} R_u T_{\text{react}} \\ P_{\text{prod}} V &= N_{\text{prod}} R_u T_{\text{prod}} \end{aligned} \right\} P_{\text{prod}} = P_{\text{react}} \left(\frac{N_{\text{prod}}}{N_{\text{react}}} \right) \left(\frac{T_{\text{prod}}}{T_{\text{react}}} \right)$$

Substituting, we get

$$P_{\text{prod}} = (1 \text{ atm}) \left(\frac{4 \text{ lbmol}}{4 \text{ lbmol}} \right) \left(\frac{1800 \text{ R}}{537 \text{ R}} \right) = \mathbf{3.35 \text{ atm}}$$

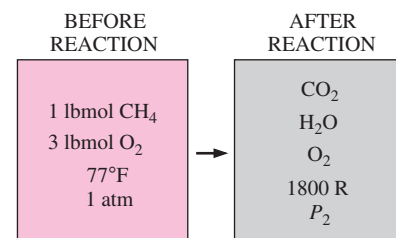


FIGURE 15-24

Schematic for Example 15-7.

(b) Noting that the process involves no work interactions, the heat transfer during this constant-volume combustion process can be determined from the energy balance $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ applied to the tank,

$$-Q_{\text{out}} = \sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v})_p - \sum N_r(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v})_r$$

Since both the reactants and the products are assumed to be ideal gases, all the internal energy and enthalpies depend on temperature only, and the $P\bar{v}$ terms in this equation can be replaced by $R_u T$. It yields

$$Q_{\text{out}} = \sum N_r(\bar{h}_f^\circ - R_u T)_r - \sum N_p(\bar{h}_f^\circ + \bar{h}_{1800\text{ R}} - \bar{h}_{537\text{ R}} - R_u T)_p$$

since the reactants are at the standard reference temperature of 537 R. From $\bar{h}_f^\circ$ and ideal-gas tables in the Appendix,

Substance	$\bar{h}_f^\circ$ Btu/lbmol	$\bar{h}_{537\text{ R}}$ Btu/lbmol	$\bar{h}_{1800\text{ R}}$ Btu/lbmol
CH ₄	-32,210	—	—
O ₂	0	3725.1	13,485.8
CO ₂	-169,300	4027.5	18,391.5
H ₂ O(g)	-104,040	4258.0	15,433.0

Substituting, we have

$$\begin{aligned} Q_{\text{out}} &= (1 \text{ lbmol CH}_4)[(-32,210 - 1.986 \times 537) \text{ Btu/lbmol CH}_4] \\ &\quad + (3 \text{ lbmol O}_2)[(0 - 1.986 \times 537) \text{ Btu/lbmol O}_2] \\ &\quad - (1 \text{ lbmol CO}_2)[(-169,300 + 18,391.5 - 4027.5 - 1.986 \times 1800) \\ &\quad \quad \text{Btu/lbmol CO}_2] \\ &\quad - (2 \text{ lbmol H}_2\text{O})[(-104,040 + 15,433.0 - 4258.0 - 1.986 \times 1800) \\ &\quad \quad \text{Btu/lbmol H}_2\text{O}] \\ &\quad - (1 \text{ lbmol O}_2)[(0 + 13,485.8 - 3725.1 - 1.986 \times 1800) \text{ Btu/lbmol O}_2] \\ &= \mathbf{308,730 \text{ Btu/lbmol CH}_4} \end{aligned}$$

Discussion On a mass basis, the heat transfer from the tank would be $308,730/16 = 19,300$ Btu/lbm of methane.

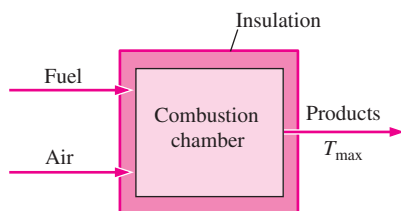


FIGURE 15–25

The temperature of a combustion chamber becomes maximum when combustion is complete and no heat is lost to the surroundings ($Q = 0$).

15–5 ■ ADIABATIC FLAME TEMPERATURE

In the absence of any work interactions and any changes in kinetic or potential energies, the chemical energy released during a combustion process either is lost as heat to the surroundings or is used internally to raise the temperature of the combustion products. The smaller the heat loss, the larger the temperature rise. In the limiting case of no heat loss to the surroundings ($Q = 0$), the temperature of the products reaches a maximum, which is called the **adiabatic flame** or **adiabatic combustion temperature** of the reaction (Fig. 15–25).

The adiabatic flame temperature of a steady-flow combustion process is determined from Eq. 15–11 by setting $Q = 0$ and $W = 0$. It yields

$$H_{\text{prod}} = H_{\text{react}} \quad (15-16)$$

or

$$\sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p = \sum N_r(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_r \quad (15-17)$$

Once the reactants and their states are specified, the enthalpy of the reactants H_{react} can be easily determined. The calculation of the enthalpy of the products H_{prod} is not so straightforward, however, because the temperature of the products is not known prior to the calculations. Therefore, the determination of the adiabatic flame temperature requires the use of an iterative technique unless equations for the sensible enthalpy changes of the combustion products are available. A temperature is assumed for the product gases, and the H_{prod} is determined for this temperature. If it is not equal to H_{react} , calculations are repeated with another temperature. The adiabatic flame temperature is then determined from these two results by interpolation. When the oxidant is air, the product gases mostly consist of N_2 , and a good first guess for the adiabatic flame temperature is obtained by treating the entire product gases as N_2 .

In combustion chambers, the highest temperature to which a material can be exposed is limited by metallurgical considerations. Therefore, the adiabatic flame temperature is an important consideration in the design of combustion chambers, gas turbines, and nozzles. The maximum temperatures that occur in these devices are considerably lower than the adiabatic flame temperature, however, since the combustion is usually incomplete, some heat loss takes place, and some combustion gases dissociate at high temperatures (Fig. 15–26). The maximum temperature in a combustion chamber can be controlled by adjusting the amount of excess air, which serves as a coolant.

Note that the adiabatic flame temperature of a fuel is not unique. Its value depends on (1) the state of the reactants, (2) the degree of completion of the reaction, and (3) the amount of air used. For a specified fuel at a specified state burned with air at a specified state, *the adiabatic flame temperature attains its maximum value when complete combustion occurs with the theoretical amount of air.*

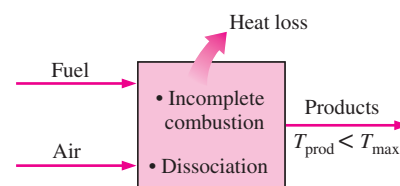


FIGURE 15–26

The maximum temperature encountered in a combustion chamber is lower than the theoretical adiabatic flame temperature.

EXAMPLE 15–8 Adiabatic Flame Temperature in Steady Combustion

Liquid octane (C_8H_{18}) enters the combustion chamber of a gas turbine steadily at 1 atm and 25°C , and it is burned with air that enters the combustion chamber at the same state, as shown in Fig. 15–27. Determine the adiabatic flame temperature for (a) complete combustion with 100 percent theoretical air, (b) complete combustion with 400 percent theoretical air, and (c) incomplete combustion (some CO in the products) with 90 percent theoretical air.

Solution Liquid octane is burned steadily. The adiabatic flame temperature is to be determined for different cases.

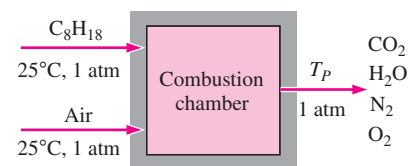


FIGURE 15–27

Schematic for Example 15–8.

Assumptions **1** This is a steady-flow combustion process. **2** The combustion chamber is adiabatic. **3** There are no work interactions. **4** Air and the combustion gases are ideal gases. **5** Changes in kinetic and potential energies are negligible.

Analysis (a) The balanced equation for the combustion process with the theoretical amount of air is



The adiabatic flame temperature relation $H_{\text{prod}} = H_{\text{react}}$ in this case reduces to

$$\sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p = \sum N_r \bar{h}_{f,r}^\circ = (N\bar{h}_f^\circ)_{\text{C}_8\text{H}_{18}}$$

since all the reactants are at the standard reference state and $\bar{h}_f^\circ = 0$ for O_2 and N_2 . The $\bar{h}_f^\circ$ and h values of various components at 298 K are

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol
$\text{C}_8\text{H}_{18}(\ell)$	-249,950	—
O_2	0	8682
N_2	0	8669
$\text{H}_2\text{O}(g)$	-241,820	9904
CO_2	-393,520	9364

Substituting, we have

$$\begin{aligned} & (8 \text{ kmol CO}_2)[(-393,520 + \bar{h}_{\text{CO}_2} - 9364) \text{ kJ/kmol CO}_2] \\ & + (9 \text{ kmol H}_2\text{O})[(-241,820 + \bar{h}_{\text{H}_2\text{O}} - 9904) \text{ kJ/kmol H}_2\text{O}] \\ & + (47 \text{ kmol N}_2)[(0 + \bar{h}_{\text{N}_2} - 8669) \text{ kJ/kmol N}_2] \\ & = (1 \text{ kmol C}_8\text{H}_{18})(-249,950 \text{ kJ/kmol C}_8\text{H}_{18}) \end{aligned}$$

which yields

$$8\bar{h}_{\text{CO}_2} + 9\bar{h}_{\text{H}_2\text{O}} + 47\bar{h}_{\text{N}_2} = 5,646,081 \text{ kJ}$$

It appears that we have one equation with three unknowns. Actually we have only one unknown—the temperature of the products T_{prod} —since $h = h(T)$ for ideal gases. Therefore, we have to use an equation solver such as EES or a trial-and-error approach to determine the temperature of the products.

A first guess is obtained by dividing the right-hand side of the equation by the total number of moles, which yields $5,646,081/(8 + 9 + 47) = 88,220$ kJ/kmol. This enthalpy value corresponds to about 2650 K for N_2 , 2100 K for H_2O , and 1800 K for CO_2 . Noting that the majority of the moles are N_2 , we see that T_{prod} should be close to 2650 K, but somewhat under it. Therefore, a good first guess is 2400 K. At this temperature,

$$\begin{aligned} 8\bar{h}_{\text{CO}_2} + 9\bar{h}_{\text{H}_2\text{O}} + 47\bar{h}_{\text{N}_2} &= 8 \times 125,152 + 9 \times 103,508 + 47 \times 79,320 \\ &= 5,660,828 \text{ kJ} \end{aligned}$$

This value is higher than 5,646,081 kJ. Therefore, the actual temperature is slightly under 2400 K. Next we choose 2350 K. It yields

$$8 \times 122,091 + 9 \times 100,846 + 47 \times 77,496 = 5,526,654$$

which is lower than 5,646,081 kJ. Therefore, the actual temperature of the products is between 2350 and 2400 K. By interpolation, it is found to be $T_{\text{prod}} = 2395 \text{ K}$.

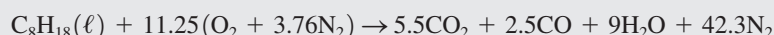
(b) The balanced equation for the complete combustion process with 400 percent theoretical air is



By following the procedure used in (a), the adiabatic flame temperature in this case is determined to be $T_{\text{prod}} = 962 \text{ K}$.

Notice that the temperature of the products decreases significantly as a result of using excess air.

(c) The balanced equation for the incomplete combustion process with 90 percent theoretical air is



Following the procedure used in (a), we find the adiabatic flame temperature in this case to be $T_{\text{prod}} = 2236 \text{ K}$.

Discussion Notice that the adiabatic flame temperature decreases as a result of incomplete combustion or using excess air. Also, *the maximum adiabatic flame temperature is achieved when complete combustion occurs with the theoretical amount of air.*

15-6 ■ ENTROPY CHANGE OF REACTING SYSTEMS

So far we have analyzed combustion processes from the conservation of mass and the conservation of energy points of view. The thermodynamic analysis of a process is not complete, however, without the examination of the second-law aspects. Of particular interest are the exergy and exergy destruction, both of which are related to entropy.

The entropy balance relations developed in Chap. 7 are equally applicable to both reacting and nonreacting systems provided that the entropies of individual constituents are evaluated properly using a common basis. The **entropy balance** for *any system* (including reacting systems) undergoing *any process* can be expressed as

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}} \quad (\text{kJ/K}) \quad (15-18)$$

Using quantities per unit mole of fuel and taking the positive direction of heat transfer to be *to* the system, the entropy balance relation can be expressed more explicitly for a *closed* or *steady-flow* reacting system as (Fig. 15-28)

$$\sum \frac{Q_k}{T_k} + S_{\text{gen}} = S_{\text{prod}} - S_{\text{react}} \quad (\text{kJ/K}) \quad (15-19)$$

where T_k is temperature at the boundary where Q_k crosses it. For an *adiabatic process* ($Q = 0$), the entropy transfer term drops out and Eq. 15-19 reduces to

$$S_{\text{gen,adiabatic}} = S_{\text{prod}} - S_{\text{react}} \geq 0 \quad (15-20)$$

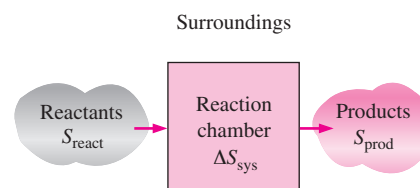


FIGURE 15-28

The entropy change associated with a chemical relation.

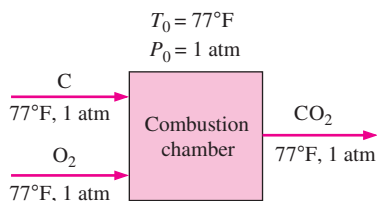


FIGURE 15-32

Schematic for Example 15-9.

EXAMPLE 15-9 Reversible Work Associated with a Combustion Process

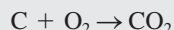
One lbmol of carbon at 77°F and 1 atm is burned steadily with 1 lbmol of oxygen at the same state as shown in Fig. 15-32. The CO₂ formed during the process is then brought to 77°F and 1 atm, the conditions of the surroundings. Assuming the combustion is complete, determine the reversible work for this process.

Solution Carbon is burned steadily with pure oxygen. The reversible work associated with this process is to be determined.

Assumptions 1 Combustion is complete. 2 Steady-flow conditions exist during combustion. 3 Oxygen and the combustion gases are ideal gases. 4 Changes in kinetic and potential energies are negligible.

Properties The Gibbs function of formation at 77°F and 1 atm is 0 for C and O₂, and -169,680 Btu/lbmol for CO₂. The enthalpy of formation is 0 for C and O₂, and -169,300 Btu/lbmol for CO₂. The absolute entropy is 1.36 Btu/lbmol · R for C, 49.00 Btu/lbmol · R for O₂, and 51.07 Btu/lbmol · R for CO₂ (Table A-26E).

Analysis The combustion equation is



The C, O₂, and CO₂ are at 77°F and 1 atm, which is the standard reference state and also the state of the surroundings. Therefore, the reversible work in this case is simply the difference between the Gibbs function of formation of the reactants and that of the products (Eq. 15-27):

$$\begin{aligned} W_{\text{rev}} &= \sum N_r \bar{g}_{f,r}^\circ - \sum N_p \bar{g}_{f,p}^\circ \\ &= N_{\text{C}} \bar{g}_{f,\text{C}}^\circ + N_{\text{O}_2} \bar{g}_{f,\text{O}_2}^\circ - N_{\text{CO}_2} \bar{g}_{f,\text{CO}_2}^\circ = -N_{\text{CO}_2} \bar{g}_{f,\text{CO}_2}^\circ \\ &= (-1 \text{ lbmol})(-169,680 \text{ Btu/lbmol}) \\ &= \mathbf{169,680 \text{ Btu}} \end{aligned}$$

since the $\bar{g}_f^\circ$ of stable elements at 77°F and 1 atm is zero. Therefore, 169,680 Btu of work could be done as 1 lbmol of C is burned with 1 lbmol of O₂ at 77°F and 1 atm in an environment at the same state. The reversible work in this case represents the exergy of the reactants since the product (the CO₂) is at the state of the surroundings.

Discussion We could also determine the reversible work without involving the Gibbs function by using Eq. 15-24:

$$\begin{aligned} W_{\text{rev}} &= \sum N_r (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - T_0 \bar{s})_r - \sum N_p (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - T_0 \bar{s})_p \\ &= \sum N_r (\bar{h}_f^\circ - T_0 \bar{s})_r - \sum N_p (\bar{h}_f^\circ - T_0 \bar{s})_p \\ &= N_{\text{C}} (\bar{h}_f^\circ - T_0 \bar{s}^\circ)_{\text{C}} + N_{\text{O}_2} (\bar{h}_f^\circ - T_0 \bar{s}^\circ)_{\text{O}_2} - N_{\text{CO}_2} (\bar{h}_f^\circ - T_0 \bar{s}^\circ)_{\text{CO}_2} \end{aligned}$$

Substituting the enthalpy of formation and absolute entropy values, we obtain

$$\begin{aligned} W_{\text{rev}} &= (1 \text{ lbmol C})[0 - (537 \text{ R})(1.36 \text{ Btu/lbmol} \cdot \text{R})] \\ &\quad + (1 \text{ lbmol O}_2)[0 - (537 \text{ R})(49.00 \text{ Btu/lbmol} \cdot \text{R})] \\ &\quad - (1 \text{ lbmol CO}_2)[-169,300 \text{ Btu/lbmol} - (537 \text{ R})(51.07 \text{ Btu/lbmol} \cdot \text{R})] \\ &= \mathbf{169,680 \text{ Btu}} \end{aligned}$$

which is identical to the result obtained before.

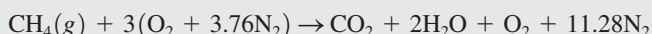
EXAMPLE 15–10 Second-Law Analysis of Adiabatic Combustion

Methane (CH_4) gas enters a steady-flow adiabatic combustion chamber at 25°C and 1 atm. It is burned with 50 percent excess air that also enters at 25°C and 1 atm, as shown in Fig. 15–33. Assuming complete combustion, determine (a) the temperature of the products, (b) the entropy generation, and (c) the reversible work and exergy destruction. Assume that $T_0 = 298\text{ K}$ and the products leave the combustion chamber at 1 atm pressure.

Solution Methane is burned with excess air in a steady-flow combustion chamber. The product temperature, entropy generated, reversible work, and exergy destroyed are to be determined.

Assumptions 1 Steady-flow conditions exist during combustion. 2 Air and the combustion gases are ideal gases. 3 Changes in kinetic and potential energies are negligible. 4 The combustion chamber is adiabatic and thus there is no heat transfer. 5 Combustion is complete.

Analysis (a) The balanced equation for the complete combustion process with 50 percent excess air is



Under steady-flow conditions, the adiabatic flame temperature is determined from $H_{\text{prod}} = H_{\text{react}}$, which reduces to

$$\sum N_p(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p = \sum N_r\bar{h}_{f,r}^\circ = (N\bar{h}_f^\circ)_{\text{CH}_4}$$

since all the reactants are at the standard reference state and $\bar{h}_f^\circ = 0$ for O_2 and N_2 . Assuming ideal-gas behavior for air and for the products, the $\bar{h}_f^\circ$ and $\bar{h}$ values of various components at 298 K can be listed as

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol
$\text{CH}_4(g)$	-74,850	—
O_2	0	8682
N_2	0	8669
$\text{H}_2\text{O}(g)$	-241,820	9904
CO_2	-393,520	9364

Substituting, we have

$$\begin{aligned} & (1 \text{ kmol CO}_2)[(-393,520 + \bar{h}_{\text{CO}_2} - 9364) \text{ kJ/kmol CO}_2] \\ & + (2 \text{ kmol H}_2\text{O})[(-241,820 + \bar{h}_{\text{H}_2\text{O}} - 9904) \text{ kJ/kmol H}_2\text{O}] \\ & + (11.28 \text{ kmol N}_2)[(0 + \bar{h}_{\text{N}_2} - 8669) \text{ kJ/kmol N}_2] \\ & + (1 \text{ kmol O}_2)[(0 + \bar{h}_{\text{O}_2} - 8682) \text{ kJ/kmol O}_2] \\ & = (1 \text{ kmol CH}_4)(-74,850 \text{ kJ/kmol CH}_4) \end{aligned}$$

which yields

$$\bar{h}_{\text{CO}_2} + 2\bar{h}_{\text{H}_2\text{O}} + \bar{h}_{\text{O}_2} + 11.28\bar{h}_{\text{N}_2} = 937,950 \text{ kJ}$$

By trial and error, the temperature of the products is found to be

$$T_{\text{prod}} = \mathbf{1789\text{ K}}$$

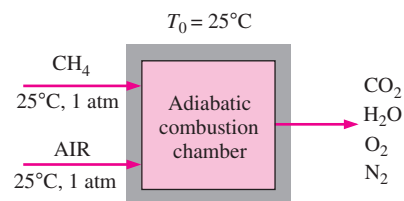


FIGURE 15–33

Schematic for Example 15–10.

(b) Noting that combustion is adiabatic, the entropy generation during this process is determined from Eq. 15–20:

$$S_{\text{gen}} = S_{\text{prod}} - S_{\text{react}} = \sum N_p \bar{s}_p - \sum N_r \bar{s}_r$$

The CH_4 is at 25°C and 1 atm, and thus its absolute entropy is $\bar{s}_{\text{CH}_4} = 186.16 \text{ kJ/kmol} \cdot \text{K}$ (Table A–26). The entropy values listed in the ideal-gas tables are for 1 atm pressure. Both the air and the product gases are at a total pressure of 1 atm, but the entropies are to be calculated at the partial pressure of the components, which is equal to $P_i = y_i P_{\text{total}}$, where y_i is the mole fraction of component i . From Eq. 15–22:

$$S_i = N_i \bar{s}_i(T, P_i) = N_i [\bar{s}_i^\circ(T, P_0) - R_u \ln y_i P_m]$$

The entropy calculations can be represented in tabular form as follows:

	N_i	y_i	$\bar{s}_i^\circ(T, 1 \text{ atm})$	$-R_u \ln y_i P_m$	$N_i \bar{s}_i$
CH_4	1	1.00	186.16	—	186.16
O_2	3	0.21	205.04	12.98	654.06
N_2	11.28	0.79	191.61	1.96	2183.47
					$S_{\text{react}} = 3023.69$
CO_2	1	0.0654	302.517	22.674	325.19
H_2O	2	0.1309	258.957	16.905	551.72
O_2	1	0.0654	264.471	22.674	287.15
N_2	11.28	0.7382	247.977	2.524	2825.65
					$S_{\text{prod}} = 3989.71$

Thus,

$$\begin{aligned} S_{\text{gen}} &= S_{\text{prod}} - S_{\text{react}} = (3989.71 - 3023.69) \text{ kJ/kmol} \cdot \text{K CH}_4 \\ &= \mathbf{966.0 \text{ kJ/kmol} \cdot \text{K}} \end{aligned}$$

(c) The exergy destruction or irreversibility associated with this process is determined from Eq. 15–23,

$$\begin{aligned} X_{\text{destroyed}} &= T_0 S_{\text{gen}} = (298 \text{ K})(966.0 \text{ kJ/kmol} \cdot \text{K}) \\ &= \mathbf{288 \text{ MJ/kmol CH}_4} \end{aligned}$$

That is, 288 MJ of work potential is wasted during this combustion process for each kmol of methane burned. This example shows that even complete combustion processes are highly irreversible.

This process involves no actual work. Therefore, the reversible work and exergy destroyed are identical:

$$W_{\text{rev}} = \mathbf{288 \text{ MJ/kmol CH}_4}$$

That is, 288 MJ of work could be done during this process but is not. Instead, the entire work potential is wasted.

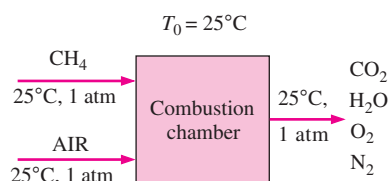


FIGURE 15–34

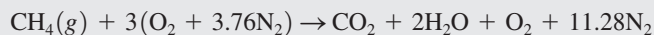
Schematic for Example 15–11.

EXAMPLE 15–11 Second-Law Analysis of Isothermal Combustion

Methane (CH_4) gas enters a steady-flow combustion chamber at 25°C and 1 atm and is burned with 50 percent excess air, which also enters at 25°C and 1 atm, as shown in Fig. 15–34. After combustion, the products are allowed to cool to 25°C . Assuming complete combustion, determine

(a) the heat transfer per kmol of CH_4 , (b) the entropy generation, and (c) the reversible work and exergy destruction. Assume that $T_0 = 298 \text{ K}$ and the products leave the combustion chamber at 1 atm pressure.

Solution This is the same combustion process we discussed in Example 15–10, except that the combustion products are brought to the state of the surroundings by transferring heat from them. Thus the combustion equation remains the same:



At 25°C , part of the water will condense. The amount of water vapor that remains in the products is determined from (see Example 15–3)

$$\frac{N_v}{N_{\text{gas}}} = \frac{P_v}{P_{\text{total}}} = \frac{3.1698 \text{ kPa}}{101.325 \text{ kPa}} = 0.03128$$

and

$$N_v = \left(\frac{P_v}{P_{\text{total}}} \right) N_{\text{gas}} = (0.03128)(13.28 + N_v) \rightarrow N_v = 0.43 \text{ kmol}$$

Therefore, 1.57 kmol of the H_2O formed is in the liquid form, which is removed at 25°C and 1 atm. When one is evaluating the partial pressures of the components in the product gases, the only water molecules that need to be considered are those that are in the vapor phase. As before, all the gaseous reactants and products are treated as ideal gases.

(a) Heat transfer during this steady-flow combustion process is determined from the steady-flow energy balance $E_{\text{out}} = E_{\text{in}}$ on the combustion chamber,

$$Q_{\text{out}} + \sum N_p \bar{h}_{f,p}^\circ = \sum N_r \bar{h}_{f,r}^\circ$$

since all the reactants and products are at the standard reference of 25°C and the enthalpy of ideal gases depends on temperature only. Solving for Q_{out} and substituting the $\bar{h}_f^\circ$ values, we have

$$\begin{aligned} Q_{\text{out}} &= (1 \text{ kmol CH}_4)(-74,850 \text{ kJ/kmol CH}_4) \\ &\quad - (1 \text{ kmol CO}_2)(-393,520 \text{ kJ/kmol CO}_2) \\ &\quad - [0.43 \text{ kmol H}_2\text{O}(g)][-241,820 \text{ kJ/kmol H}_2\text{O}(g)] \\ &\quad - [1.57 \text{ kmol H}_2\text{O}(\ell)][-285.830 \text{ kJ/kmol H}_2\text{O}(\ell)] \\ &= \mathbf{871,400 \text{ kJ/kmol CH}_4} \end{aligned}$$

(b) The entropy of the reactants was evaluated in Example 15–10 and was determined to be $S_{\text{react}} = 3023.69 \text{ kJ/kmol} \cdot \text{K CH}_4$. By following a similar approach, the entropy of the products is determined to be

	N_i	y_i	$\bar{s}_i^\circ(T, 1 \text{ atm})$	$-R_u \ln y_i P_m$	$N_i \bar{s}_i$
$\text{H}_2\text{O}(\ell)$	1.57	1.0000	69.92	—	109.77
H_2O	0.43	0.0314	188.83	28.77	93.57
CO_2	1	0.0729	213.80	21.77	235.57
O_2	1	0.0729	205.04	21.77	226.81
N_2	11.28	0.8228	191.61	1.62	2179.63
$S_{\text{prod}} = 2845.35$					

Then the total entropy generation during this process is determined from an entropy balance applied on an *extended system* that includes the immediate surroundings of the combustion chamber

$$\begin{aligned} S_{\text{gen}} &= S_{\text{prod}} - S_{\text{react}} + \frac{Q_{\text{out}}}{T_{\text{surr}}} \\ &= (2845.35 - 3023.69) \text{ kJ/kmol} + \frac{871,400 \text{ kJ/kmol}}{298 \text{ K}} \\ &= \mathbf{2746 \text{ kJ/kmol} \cdot \text{K CH}_4} \end{aligned}$$

(c) The exergy destruction and reversible work associated with this process are determined from

$$\begin{aligned} X_{\text{destroyed}} &= T_0 S_{\text{gen}} = (298 \text{ K})(2746 \text{ kJ/kmol} \cdot \text{K}) \\ &= \mathbf{818 \text{ MJ/kmol CH}_4} \end{aligned}$$

and

$$W_{\text{rev}} = X_{\text{destroyed}} = \mathbf{818 \text{ MJ/kmol CH}_4}$$

since this process involves no actual work. Therefore, 818 MJ of work could be done during this process but is not. Instead, the entire work potential is wasted. The reversible work in this case represents the exergy of the reactants before the reaction starts since the products are in equilibrium with the surroundings, that is, they are at the dead state.

Discussion Note that, for simplicity, we calculated the entropy of the product gases before they actually entered the atmosphere and mixed with the atmospheric gases. A more complete analysis would consider the composition of the atmosphere and the mixing of the product gases with the gases in the atmosphere, forming a homogeneous mixture. There is additional entropy generation during this mixing process, and thus additional wasted work potential.

TOPIC OF SPECIAL INTEREST*

Fuel Cells

Fuels like methane are commonly burned to provide thermal energy at high temperatures for use in heat engines. However, a comparison of the reversible works obtained in the last two examples reveals that the exergy of the reactants (818 MJ/kmol CH₄) decreases by 288 MJ/kmol as a result of the irreversible adiabatic combustion process alone. That is, the exergy of the hot combustion gases at the end of the adiabatic combustion process is 818 – 288 = 530 MJ/kmol CH₄. In other words, the work potential of the hot combustion gases is about 65 percent of the work potential of the reactants. It seems that when methane is burned, 35 percent of the work potential is lost before we even start using the thermal energy (Fig. 15–35).

Thus, the second law of thermodynamics suggests that there should be a better way of converting the chemical energy to work. The better way is, of course, the less irreversible way, the best being the reversible case. In chemi-

*This section can be skipped without a loss in continuity.